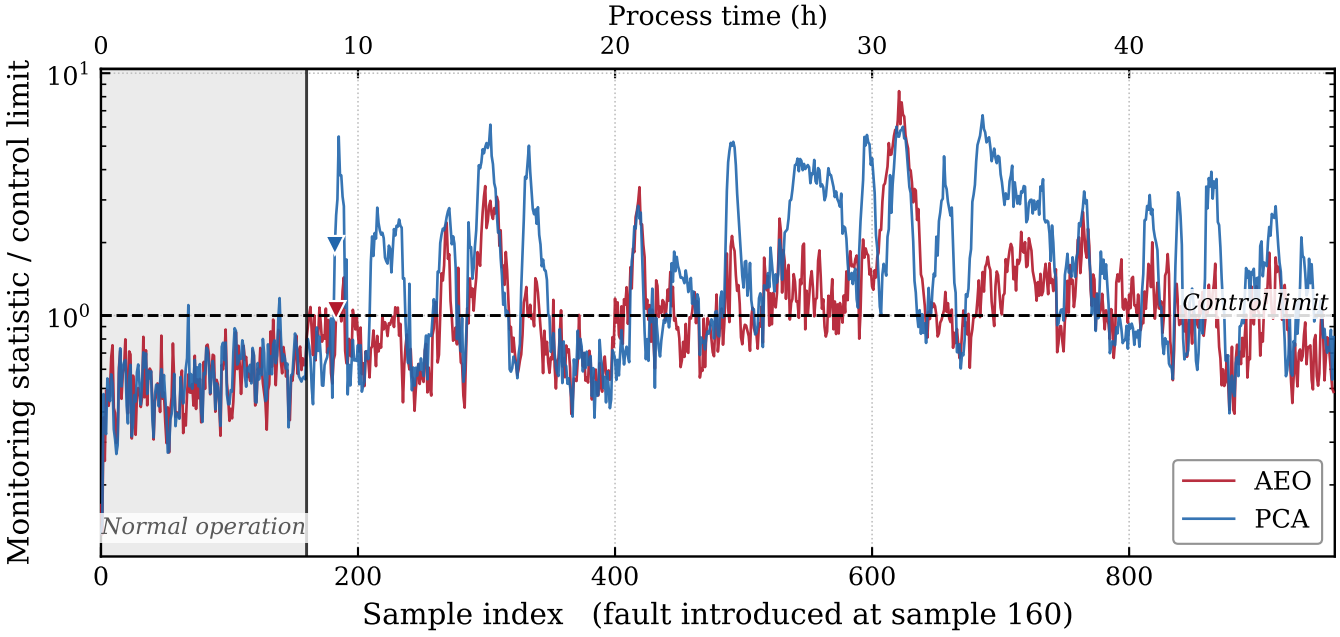
Fault Detection Performance for IDV(10)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	71.6 samples (215 min)	MDR 0.529	detected in 7/7 runs
PCA	delay	36.9 samples (111 min)	MDR 0.346	detected in 7/7 runs

PCA detects 34.7 samples (104 min) earlier; PCA missed-detection rate lower by 35%